

Supplementary Material

Transmission Expansion and Reliability in Peru's Energy Transition: A System Dynamics Approach

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S1 Model equations

This section states the model’s core relationships. The full equation set, in executable form, is in the Vensim model distributed with the paper; what follows is the subset a reader needs in order to judge the structure.

S1.1 Sets and indices

$z \in \mathcal{Z}$	zone; three active (Z1 north, Z2 centre, Z3 south) of six represented
$g \in \mathcal{G}$	generation technology; fourteen per zone, eleven with non-zero potential
$i \in \mathcal{I}$	interconnection corridor; two active (Z1–Z2, Z2–Z3) of fifteen possible pairs
$h \in \mathcal{H}$	hour of the representative day, $ \mathcal{H} = 24$
$f \in \mathcal{F}$	forecast horizon, $\{f_0, f_1, f_3, f_5, f_7, f_9\}$ years ahead
t	time, months elapsed from January 2025; horizon 300 months, step 0.25 month

S1.2 Generation pipeline

Committed capacity flows through licensing and construction before commissioning:

$$\dot{Q}_{z,g}^{\text{lic}} = N_{z,g}(t) - \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}}](t) \quad (1)$$

$$\dot{Q}_{z,g}^{\text{con}} = \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}}](t) - \mathcal{D}[N_{z,g}, L_{z,g}^{\text{lic}} + L_{z,g}^{\text{con}}](t) \quad (2)$$

$$\dot{K}_{z,g} = \min\{\mathcal{D}[N_{z,g}, L_{z,g}^{\text{tot}}](t), \Phi_z(t)\} \quad (3)$$

with $\mathcal{D}[\cdot, L]$ a material delay of duration L and

$$L_{z,g}^{\text{tot}} = (L_{z,g}^{\text{lic}} + L_{z,g}^{\text{con}})(1 + \Delta_g). \quad (4)$$

$\Phi_z(t)$ is the commissioning headroom the delivered network allows in zone z . The sensitivity knob Δ_g replaces the workbook’s “other delays” fraction; its base value for Peru is -0.20 , so a change δ enters as $\Delta_g = (1 - 0.20)(1 + \delta) - 1$.

S1.3 Investment trigger

Two mechanisms generate $N_{z,g}$. The adequacy mechanism converts a shortfall against the desired margin into a requirement:

$$R_f^{\text{adq}}(t) = \max\left\{0, (1 + \bar{m}) \hat{D}_f(t) - \sum_{z,g} \tilde{K}_{z,g,f}(t)\right\} \quad (5)$$

where $\hat{D}_f$ is peak demand at horizon f and $\tilde{K}_{z,g,f}$ is expected capacity at that horizon, including pipeline. The profitability mechanism admits options whose profitability index exceeds Π^* :

$$\text{NPV}_{z,g}(t) = \sum_{y=1}^{25} \frac{\mathcal{R}_{z,g}(t, y) - \mathcal{C}_{z,g}(t, y)}{(1 + r)^y} - I_{z,g}(t), \quad \Pi_{z,g} = 1 + \frac{\text{NPV}_{z,g}}{I_{z,g}} \geq \Pi^*. \quad (6)$$

Admitted options are allocated across technologies by a logit share

$$s_{z,g} = \frac{\exp(\text{NPV}_{z,g}/\theta)}{\sum_{g' \in \mathcal{G}^+} \exp(\text{NPV}_{z,g'}/\theta)}, \quad (7)$$

with $\mathcal{G}^+$ the admitted set. The scale θ represents dispersion of investor beliefs: as $\theta \rightarrow 0$ the allocation collapses onto the single best option, recovering an optimising agent; as $\theta \rightarrow \infty$ it becomes uniform.

S1.4 Transmission

$$T_z^d(t) = \kappa_d \hat{D}_z(t) \quad (\text{demand connection, no delay}) \quad (8)$$

$$\dot{T}_z^g(t) = \kappa_g \mathcal{D}[N_z, L^{\text{tot},T}] (t) \quad (\text{generation connection}) \quad (9)$$

$$\dot{P}_i(t) = \mathcal{D}[\Delta P_i, L_i^{\text{tot},T}] (t) \quad (\text{interconnection}) \quad (10)$$

with the corridor increment triggered by the unconstrained dispatch,

$$\Delta P_i(t) = \text{clip}((1 + \sigma) \Theta_i(t) - P_i(t), F_{\min}, F_{\max}), \quad (11)$$

where Θ_i is the transfer the unconstrained dispatch would route through corridor i and σ the grid security margin. If the shortfall is at most $\alpha P_i(t)$ it may be met by a grid-enhancing technology, which uses the shorter GET pipeline. Transmission lead times are $L^{\text{tot},T} = \sum_s L_s (1 + \Delta_t)$ over stages $s \in \{\text{tender, investor selection, licensing, construction}\}$; the 12-month planning cycle is excluded from the scaling.

S1.5 Dispatch and price

For each month the model dispatches 24 hourly levels by merit order on offer price, first unconstrained (to size Θ_i) and then subject to $P_i(t)$. Reservoir hydro offers as a function of storage level between a minimum and a scarcity price, which is what gives hydrology a price rather than only an energy effect. The reported price is

$$\bar{\lambda} = \frac{1}{12} \sum_{m=T-11}^T \frac{1}{|\mathcal{Z}^a|} \sum_{z \in \mathcal{Z}^a} \lambda_z^{\text{reg}}(m), \quad (12)$$

the mean over the final twelve months of the horizon across active zones. The twelve-month window is used because the price oscillates month to month with hydrology and a single month is not representative.

S2 Parameters

All values are those of the distributed input workbook `PERU.xlsx`. Cell references are given so that every number can be located in the source.

S2.1 System-level parameters

Parameter	Cell	Value	Unit
Transmission factor per demand (κ_d)	General Parameters!B2	0.80	km MW ⁻¹
Initial transmission for generation	B3	5,622.82	km
Initial transmission for demand	B4	8,718.08	km
Old transmission correction	B5	2,413.38	km
Transmission cost (c_t)	B6	176.607	USD km ⁻¹ -day
Grid security margin (σ)	B7	0.30	–
Minimum grid increment ($F_{\min}$)	B8	200	MW
Maximum grid increment ($F_{\max}$)	B9	500	MW
Initial transmission for interconnection	B10	2,792.10	km
Desired availability margin ($\bar{m}$)	B11	0.25	–
Allowed GET share (α)	Interconnections!S3:S17	0.30	–
Profitability threshold (Π^*)	model constant	1.20	–

Parameter	Cell	Value	Unit
Logit scale (θ)	model constant	1×10^8	USD
Discount rate (r)	Gen Inv. param.!C23:CH23	0.10	per year
Technical lifetime (L_g)	Gen Inv. param.!C2:CH2	300	months

Table S1: System-level parameters.

S2.2 Configuration switches

Every run reported in the paper uses the configuration below. It is the configuration of the distributed workbook; the experiment quantifies uncertainty *around* it and does not vary it.

Switch	Cell	Value	Meaning
Zonal prices	B1	1	zonal, not single national price
Planned generation	B12	1	exogenous expansion plan active
Endogenous margin generation	B13	0	margin trigger not exclusive
DER expansion	B14	1	distributed resources endogenous
Planned DER expansion	B15	0	no exogenous DER plan
Only margin investments	B16	0	profitability trigger active
Only RES investment (margin)	B17	0	thermal admissible
Only RES new investments	B18	0	thermal admissible
Only power allocation	B19	0	energy and power
Grid-enhancing technologies	B22	0	disabled
Battery storage (system level)	B23	0	disabled
Dispatch path	B24	0	internal Vensim dispatch, not PyPSA

Table S2: Configuration switches. The two disabled options bound the interpretation of the results: the study quantifies uncertainty around the current policy configuration, not the effect of deploying storage or GETs.

S2.3 Technology parameters

Technology	Constr. [mo]	Licens. [mo]	Other [mo]	FOM [USD MW ⁻¹ -yr]	Fuel [USD MW ⁻¹ h]	ENFICC [-]
Biomass	36	24	-12.0	163,600	73.57	0.75
Wind, onshore	36	24	-12.0	27,873	0	0.20
Wind, offshore	36	24	-12.0	76,000	0	0.20
Solar PV	24	24	-9.6	16,000	0	0.05
Hydro, run-of-river	36	24	-12.0	35,000	0	0.35
Hydro, reservoir	60	24	-16.8	92,000	0	0.55
Thermal 1	36	24	-12.0	159,000	30.00	0.85
Thermal 2	36	24	-12.0	36,000	25.00	0.85
Geothermal	36	24	-12.0	143,000	0	0.85
Solar PV + BESS (1)	24	24	-9.6	52,000	0	0.40
Solar PV + BESS (2)	24	24	-9.6	52,000	0	0.40

Table S3: Generation technology parameters. “Other” is the base-case delay fraction expressed in months; it is negative because the Peruvian data set assumes realised lead times at 80 % of planned. Variable cost (not shown) is 5.24 USD MW⁻¹ h for biomass, 15.69 for Thermal 1 and 2.49 for Thermal 2, zero elsewhere. CAPEX follows a declining monthly trajectory from National Renewable Energy Laboratory (2024) and is given in full in the workbook (Gen Inv. param.!C62:CH373).

S2.4 Transmission stage times

Stage	Lines [months]	GETs [months]
Expansion plan development (not scaled)	12	12
Public tender preparation	20	20
Investor selection	9	9
Licensing	28	12
Construction	28	20
Total planned	97	73

Table S4: Transmission pipeline stage times, base case. The lead-time factor Δ_t scales every stage except the planning cycle, which the model uses as the period of a modulo operation setting when plans are revised.

S2.5 Zonal and corridor data

Corridor	Zones	Length [km]	Initial network [km]	Initial capacity [MW]
Int1	Z1–Z2	232.3	773.07	900
Int6	Z2–Z3	271.8	1,063.40	700
Int2	Z1–Z3	504.1	–	0 (no direct link)

Table S5: Interconnection corridors. Fifteen zone pairs are representable; two carry initial capacity. The absence of a direct Z1–Z3 link is the series topology discussed in Section 4 of the paper.

S3 System dynamics validation

Confidence in a system dynamics model is established by a documented battery of tests rather than by a single goodness-of-fit statistic, because the model is used to explain behaviour arising from structure rather than to forecast (Barlas, 1996; Oliva, 2004; Richardson, 2024). We report the tests in the order Barlas sets out: structure first, behaviour second, and behaviour tests only once the structure has been accepted.

S3.1 Structure verification

Every feedback loop in the model is matched to a decision process documented for the Peruvian or an analogous system, with its source. The two loops that generate the behaviour of interest are:

- B1 — Adequacy balancing.** Capacity shortfall against the desired margin $\bar{m} \rightarrow$ investment requirement $\rightarrow$ pipeline $\rightarrow$ installed capacity $\rightarrow$ margin. Delay: L_g^{tot} , 24–84 months. This is the loop whose delay produces capacity cycles (Ford, 2001; Assili et al., 2008).
- R1 — Price-driven investment.** Zonal price $\rightarrow$ NPV $\rightarrow$ admitted options $\rightarrow$ capacity $\rightarrow$ merit-order displacement $\rightarrow$ price. Reinforcing in the downward direction once low-marginal-cost capacity enters, which is the mechanism behind the price result of Section 5.4 of the paper.
- B2 — Network constraint on commissioning.** Completed capacity $\rightarrow$ requires evacuation network $\rightarrow$ transmission pipeline (97 months for lines) $\rightarrow$ commissioning headroom $\Phi_z \rightarrow$ installed capacity. The asymmetry between this delay and L_g^{tot} is the structural bottleneck the paper studies.

S3.2 Parameter verification

Every parameter is reported in Section S2 with its value, unit and source cell. Technology cost and performance parameters are from National Renewable Energy Laboratory (2024); network intensities κ_d , κ_g are empirical regressions on historical line additions classified by purpose; stage times are from the Peruvian transmission planning process.

S3.3 Dimensional consistency

The model passes Vensim’s *Units Check* with no errors. This is a necessary condition, not a sufficient one, but a model that fails it cannot be trusted at all.

S3.4 Extreme-condition tests

Six tests, each with the behaviour that structural soundness requires and the behaviour observed.

Test	Required behaviour	Observed
Zero demand growth	No capacity additions beyond the exogenous plan; network grows only with the committed plan	Confirmed: at g_D driven to its lower bound, installed capacity in 2050 falls to 48.5 GW, essentially the exogenous plan
Zero construction and licensing time	Capacity responds within one step; margin never breaches $\bar{m}$	Confirmed in the validation configuration; the margin is held and the price falls immediately, then recovers as the market rebalances
Unbounded lead times	Pipeline accumulates without commissioning; margin collapses; price rises to the scarcity price	Confirmed
Zero initial capacity	Immediate and large investment signal; no negative stocks	Confirmed; all stocks remain non-negative
Corridor capacity set to zero	Zones dispatch in isolation; deficit zones price at scarcity; interconnection trigger fires at maximum increment	Confirmed
Fuel price driven to zero	Thermal plant moves to the base of the merit order; renewable investment signal weakens but does not reverse	Confirmed; this is the structural reason fuel price ranks last in the design

Table S6: Extreme-condition tests.

S3.5 Behaviour reproduction

The central case reproduces the reference trajectory of the Peruvian data set: 63.85 GW of installed capacity and 46,341 km of transmission at January 2050, from 14,057 MW and 16,180 km at January 2025.

A stronger check is available because the experiment required the model to be re-published with an additional sensitivity constant. Re-simulating the central case on the re-published model, with all four knobs at their base values, returns 63.85 GW and 46,341.23 km — identical to the cached central run and to the reference. Since each knob enters as a multiplier whose default

value is the identity, this is the end-to-end confirmation that adding the instrumentation did not perturb the model. The check is scripted (`tools/verify_base_case.py`) and can be re-run.

S3.6 Behaviour sensitivity

The designed experiment of the paper is itself the behaviour-sensitivity test. It establishes that the qualitative conclusion — renewable-dominated expansion driven by profitability, with transmission as the binding constraint — holds across all 120 Monte Carlo samples, while the quantitative outcome is governed by demand growth. A model whose conclusion reversed within the sampled ranges would not support the policy reading in Section 6 of the paper.

S4 Experimental design

S4.1 Sampling and diagnostics

The Monte Carlo design is a Latin hypercube of 120 samples over four factors, generated with a fixed seed and written to disk before any simulation (McKay, 1992). Figure S1 shows the realised design projected onto each pair of factors, with the central and one-at-a-time runs overlaid.

Diagnostics of the realised design:

- Stratum coverage: 117 of 120 strata occupied in the least-covered factor. The three missing strata are a rounding artefact — sampled values are stored to four decimals so that a completed run can be matched to its design row by exact comparison, and a value drawn close to a stratum boundary can round across it.
- Maximum absolute correlation between factors: 0.181; mean 0.076. Near-orthogonality is what makes the rank correlations of Table 3 in the paper attributable to individual factors.

S4.2 Convergence of the reported percentiles

Figure S2 recomputes P10, P50 and P90 from growing subsets of the Monte Carlo runs. All three are stable from roughly 75 runs onwards for every indicator, which is the evidence that 120 samples suffice. This is a check, not an assumption: had the percentiles still been moving at 120, the correct response would have been to report that rather than to stop.

S4.3 Factor influence

Figure S3 plots each indicator against each factor over the Monte Carlo runs, with the Spearman rank correlation. The near-vertical scatter for fuel price and the two lead-time factors, against the tight monotone relationship for demand growth, is the visual form of Table 3 in the paper.

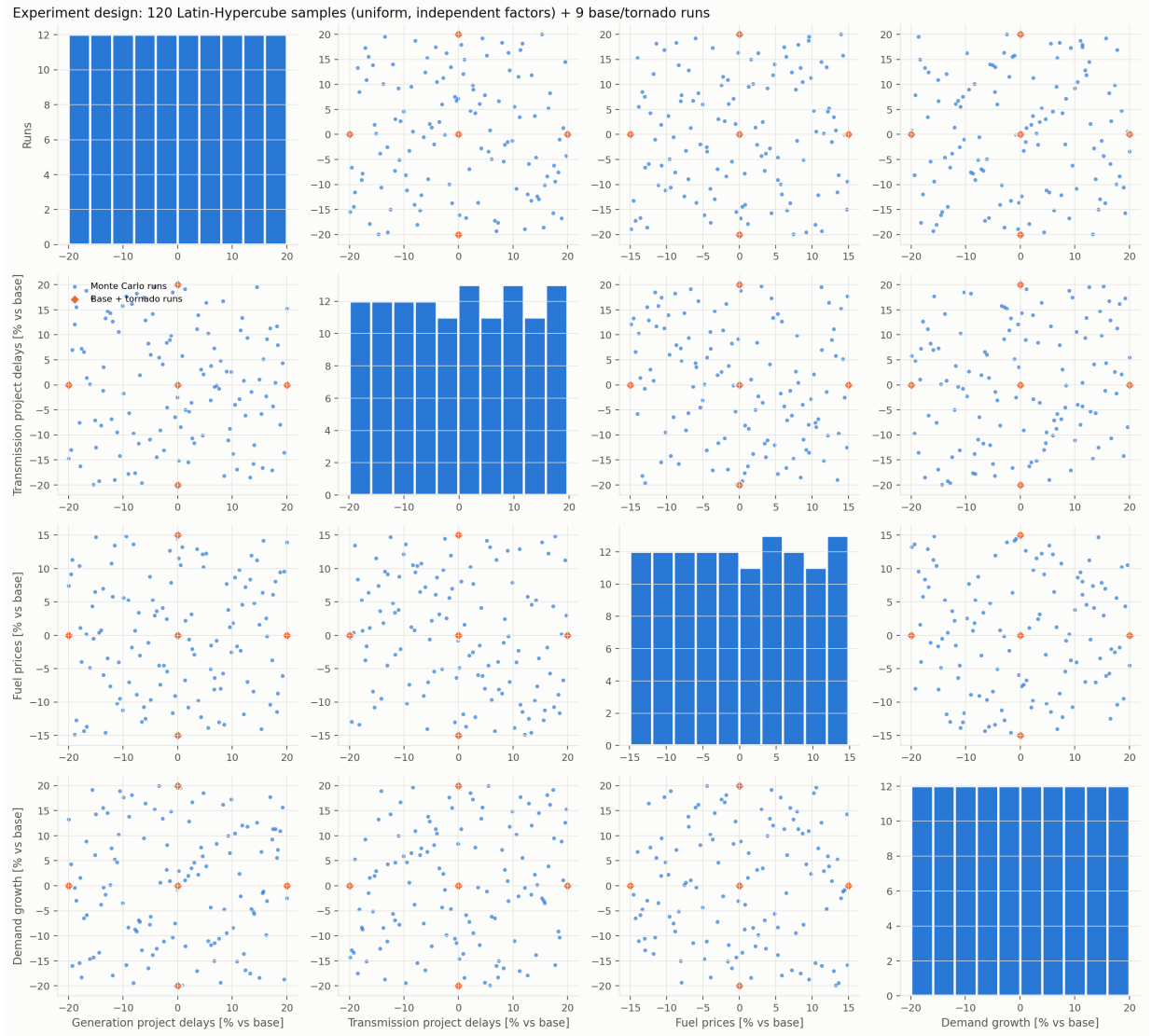


Figure S1: Experimental design. Diagonal panels are the marginal distribution of each factor over the Monte Carlo runs; off-diagonal panels are the pairwise projections. Diamonds are the central and one-at-a-time runs.

Convergence check: percentiles at 2050 recomputed as runs are added

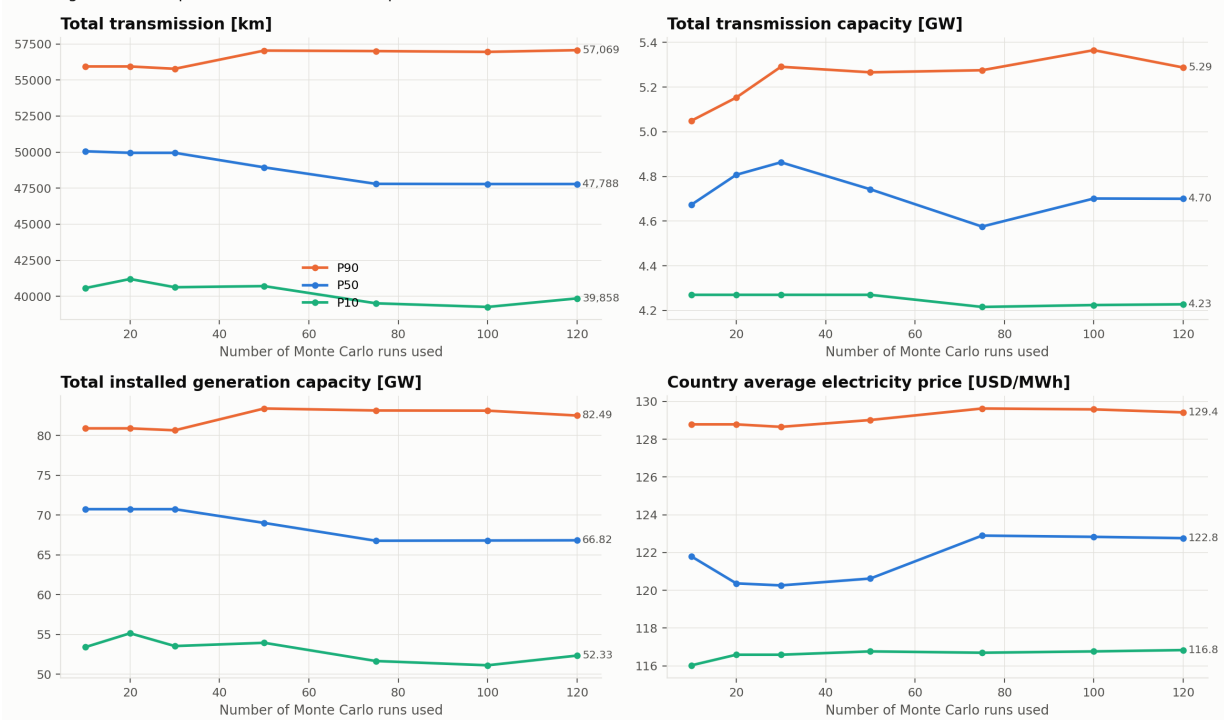


Figure S2: Convergence of the 2050 percentiles as Monte Carlo runs are added.

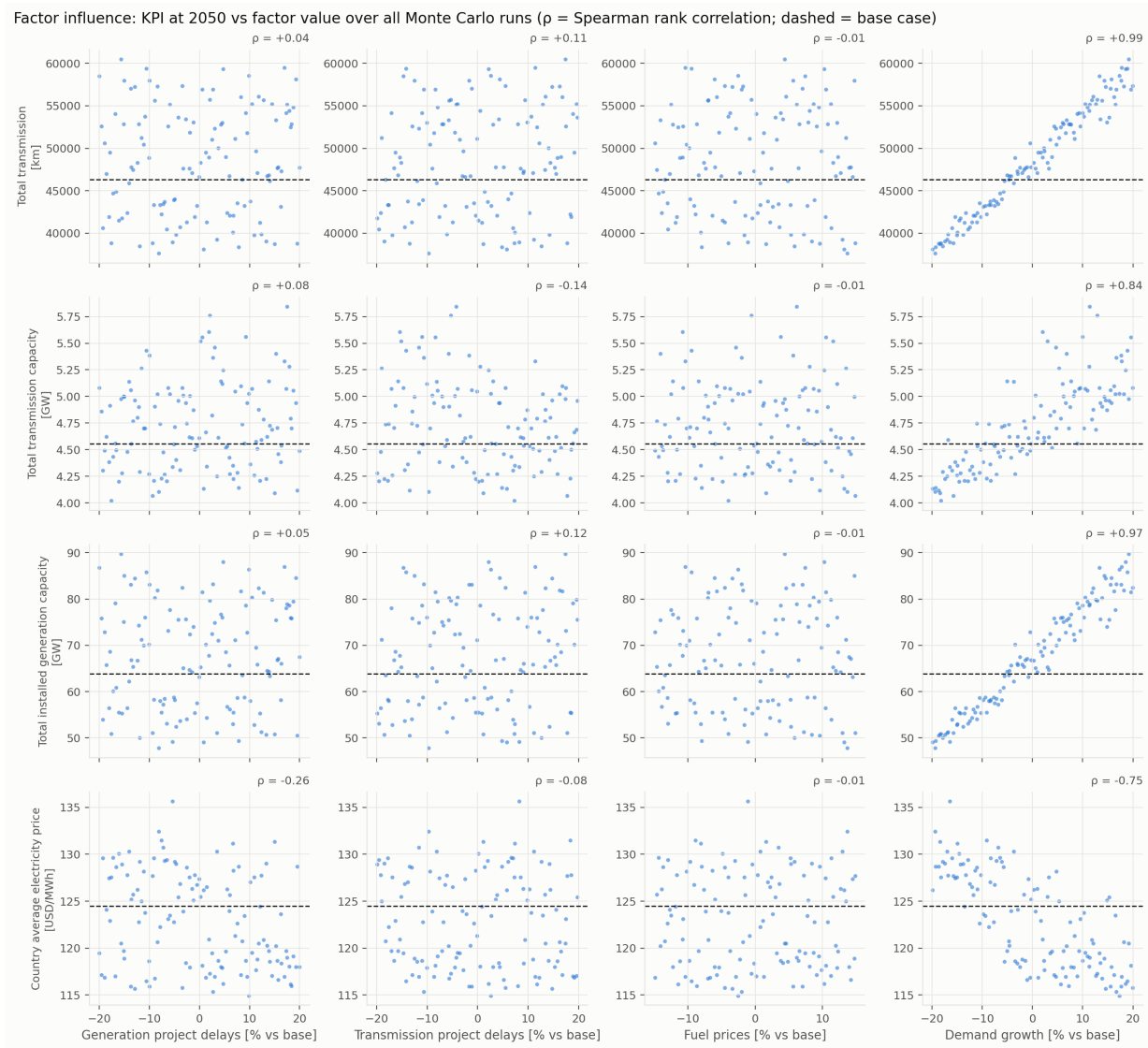


Figure S3: Indicator at 2050 against factor value over the Monte Carlo runs; ρ is the Spearman rank correlation and the dashed line is the central case.

S5 Extended results

S5.1 Distribution of outcomes in 2050

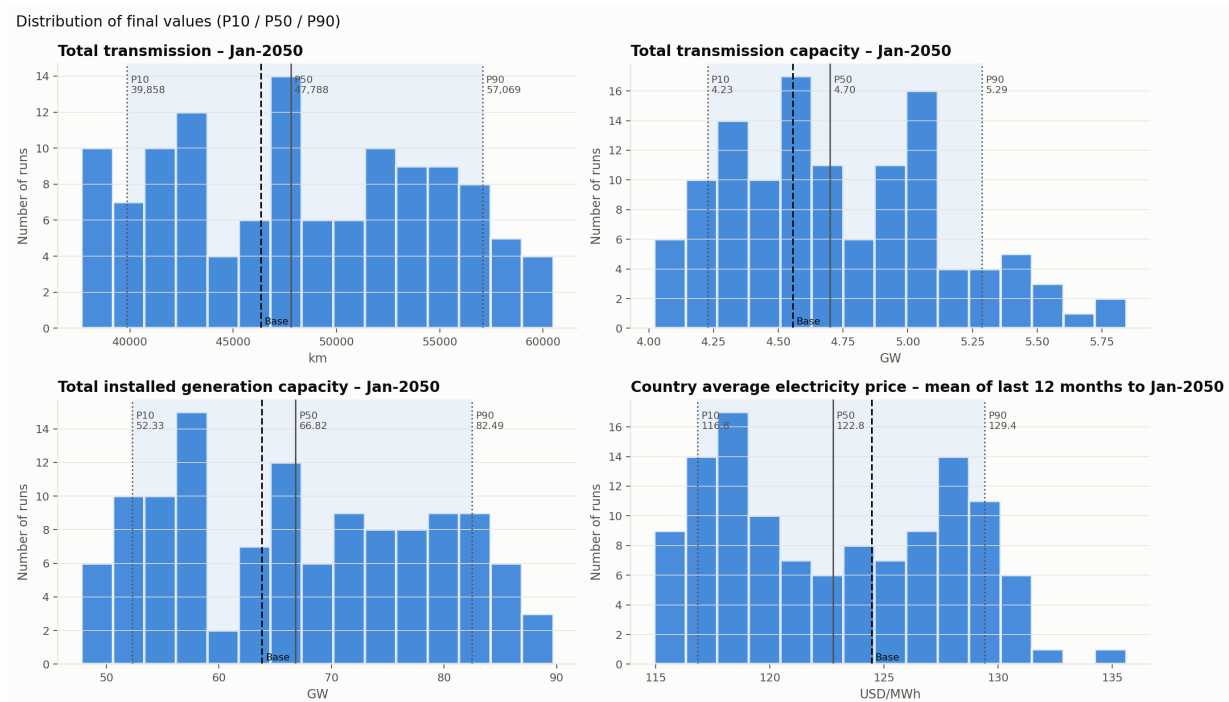


Figure S4: Distribution of each indicator at January 2050 over the 120 Monte Carlo runs, with P10, P50 and P90 marked.

S5.2 Interconnection capability

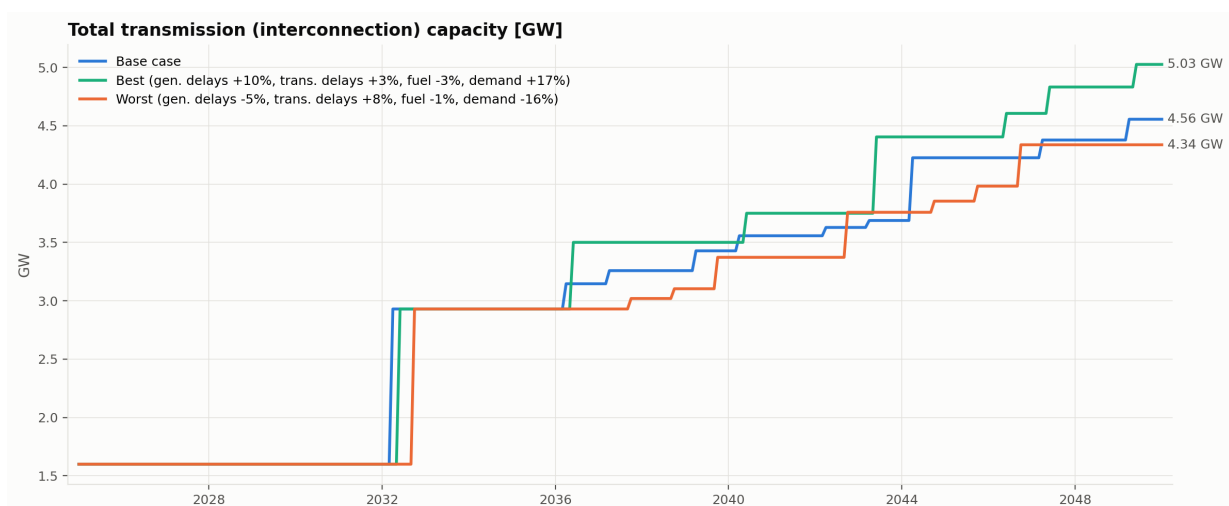


Figure S5: Inter-zonal transfer capability for the central, best and worst case.

S5.3 Technology grouping

The model's fourteen technologies per zone are aggregated into eight series for the stacked figures, because a stacked chart ceases to be readable beyond eight bands. The mapping is fixed, so a

technology keeps its colour across every figure and every scenario.

Plotted series	Model technologies
Hydro (reservoir)	THydroDam
Hydro (run-of-river)	THRoR
Thermal 1	TThermo1
Thermal 2	TThermo2
Onshore wind	TWindOnS
Solar PV	TPV
Solar PV + BESS	TPVBESSOp1, TPVBESSOp2
Other	TBiomass, TGeothermal, TWindOffS, DER

Table S7: Technology grouping used in the stacked figures.

S5.4 Technology mix of the three reported cases

Technology	Capacity 2050 [GW]			Generation [TW hr ⁻¹]		
	Central	Best	Worst	Central	Best	Worst
Hydro (reservoir)	4.87	5.30	4.57	21.6	17.6	20.8
Hydro (run-of-river)	1.62	1.62	1.61	6.5	6.1	6.8
Thermal 1	2.62	2.62	2.62	0.0	0.0	0.0
Thermal 2	7.03	7.71	6.39	7.5	5.6	10.2
Onshore wind	34.83	52.05	25.47	82.9	109.5	63.7
Solar PV	8.33	11.51	5.24	12.2	14.1	8.2
Solar PV + BESS	1.79	1.23	1.31	1.1	0.5	1.0
Other	2.76	4.35	1.96	16.8	20.0	14.7
Total	63.85	86.40	49.18	148.8	173.4	125.4
Non-thermal share	84.9 %	88.0 %	81.7 %			

Table S8: Technology mix at January 2050. Thermal 1 holds capacity but is not dispatched in the representative day by 2050 in any of the three cases, which is why its generation is zero while its capacity is non-zero.

S5.5 Summary table

Summary - KPIs at 2050 (price: mean of the last 12 months)

KPI (2050)	Base	P10	P50	P90	Best scenario	Worst scenario
Total transmission [km]	46,341	39,858	47,788	57,069	58,562 (+26.4%)	38,927 (-16.0%)
Total transmission capacity [GW]	4.56	4.23	4.70	5.29	5.03 (+10.3%)	4.34 (-4.8%)
Total installed generation capacity [GW]	63.85	52.33	66.82	82.49	86.40 (+35.3%)	49.18 (-23.0%)
Country average electricity price [USD/MWh]	124.5	116.8	122.8	129.4	114.9 (-7.6%)	135.6 (+9.0%)

Factor values of each scenario

Factor (change vs base)	Base	Best scenario	Worst scenario
Generation project delays	+0.0%	+9.8%	-5.4%
Transmission project delays	+0.0%	+2.6%	+8.2%
Fuel prices	+0.0%	-2.5%	-1.1%
Demand growth	+0.0%	+17.3%	-16.5%

Best / worst = Monte Carlo runs with lowest / highest country average price. P10-P90 from Latin-Hypercube sampling. Details in tornado_summary.xlsx (sheet README).

Figure S6: Summary of indicators and factor values for the central, best and worst case, with the ensemble percentiles.

S6 Reproducibility

The complete artefact — Vensim model, input workbook, experimental design, simulation driver, analysis code and a commented notebook that regenerates every figure — is available at the repository given in the paper. Three features of the execution are worth recording because they affected the results.

1. *The design is fixed before execution.* The Latin hypercube is written to disk before the first simulation, so the experiment is specified independently of how or whether it was run.
2. *Execution is resumable and strict about it.* Each run stores its results alongside the factor values that produced them, and a cached run is reused only if those values match the design exactly. A change to the design therefore invalidates precisely the runs it affects rather than silently mixing two experiments.
3. *A run that produces no data fails rather than being recorded.* During execution, ten runs completed without the simulation engine returning any series. Without an explicit check these would have been written to disk as empty results and counted as complete, entering the percentile calculations as 110 samples presented as 120. The driver now raises in that situation, and the ten runs were re-simulated.

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